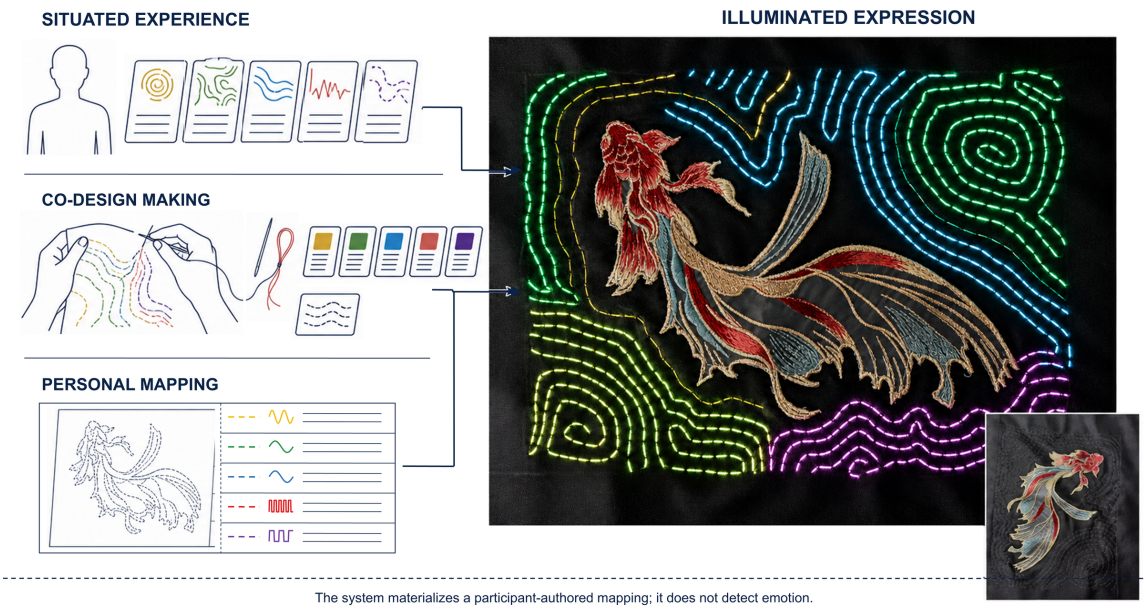
Stitching Affective Light: A Co-Design Method for Personalizable Illuminated Embroidery

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Stitching Affective Light reframes affective interface design by shifting representational authority from designer-authored emotion labels toward participant-authored material and computational expressions. We present a six-stage co-design method spanning expressive boundaries, personal affect repertoires, motif composition, placement, color-temporality authoring, and material enactment. Embroidery provides a persistent tactile layer, while programmable luminous line provides a transient layer. We articulate the method through artifact-centered Research through Design and RippleGlow, a researcher-authored worked example in which a goldfish remains stable while a linear ocean changes color and rhythm. We contribute a reproducible procedure and facilitation toolkit, a modular expressive grammar, a material-computational implementation pattern, and five design propositions concerning interpretive authority, persistent and transient expression, ambiguity, craft access, and selective disclosure. No participant deployment is reported; claims concern the designed method and worked enactment rather than usability, inclusiveness, or communicative efficacy.

CCS Concepts: • **Human-centered computing** → Participatory design; Interaction design theory, concepts and paradigms; Interaction devices.

Additional Key Words and Phrases: co-design, participatory making, illuminated embroidery, e-textiles, affective interaction, emotional expression, temporal light, personal meaning

1 INTRODUCTION

Affective technologies often promise to make emotion visible. That promise can be useful, but it can also narrow emotion into a small set of categories selected in advance by a system designer. Emotion in everyday interaction is less stable: it is culturally situated, temporally changing, and jointly interpreted through words, silence, gesture, clothing, relationship, and place [1, 2]. A red signal might mean anger, urgency, excitement, attraction, or simply a person's preferred color. The design problem is therefore not only how to display an emotion, but who has the authority to decide what the display means.

We investigate this problem through illuminated embroidery. Embroidery carries imagery, texture, labor, memory, and craft identity even when no electronics are active. Programmable light can temporarily alter that material composition through color, brightness, rhythm, and duration. Their combination creates a particularly useful tension: the stitched form can remain personally recognizable while illumination changes its situated affective reading.

Our central contribution is not a single goldfish embroidery but a method that shifts affective interface design from choosing how a system should represent emotion to arranging how people can author representations with material and computational support. Stitching Affective Light enables a person to decide what to embroider, where the piece may be worn or placed, which colors belong to their own affective vocabulary, how light should change over time, and what should remain ambiguous or private. Designers and facilitators contribute material knowledge and technical support, but do not retain exclusive control over representation. This orientation follows co-design's shift from designing for people toward enabling collective creativity and participant authorship [3, 4].

We developed the method through an artifact-centered Research through Design process. We synthesized concepts from participatory design, craft and e-textiles, interactional accounts of emotion, color research, and temporal materiality; converted them into workshop stages and tangible design resources; and then conducted a designer walkthrough using one completed example, RippleGlow. RippleGlow depicts one goldfish within a linear ocean. It is evidence that the dimensions can be composed into a functioning artifact, not evidence that other people would choose the same motif or interpret it in the same way.

We ask three research questions:

1. RQ1. How can a co-design process translate personally situated affective experience into an embroidered motif and programmable light behavior without imposing a universal emotion code?
2. RQ2. What materials, editable representations, and facilitation roles are required to preserve participant authorship across textile craft and electronic implementation, including after facilitated making ends?
3. RQ3. What design tensions become visible when the proposed method is enacted through a complete illuminated-embroidery example?

This paper contributes:

(1) Stitching Affective Light, a six-stage co-design method with explicit inputs, activities, facilitator responsibilities, participant decisions, and outputs;

(2) a modular expressive grammar that treats motif, placement, color, and temporal light as separable but interdependent dimensions;

(3) a material-computational pattern that separates participant-authored mappings, co-made textile assembly, and specialist electronic infrastructure; and

(4) a worked example and five design propositions concerning interpretive authority, persistent and transient expression, ambiguity, craft access, and selective disclosure.

2 CONTEXT AND GUIDING PRINCIPLES

2.1 Emotion as Interaction Rather Than Data

The circumplex model organizes affect through valence and arousal and remains a useful way to distinguish emotional quality from activation [1]. However, using such a model as a design scaffold is not the same as treating emotion as a fixed, irreducible data object. Boehner et al. argue for emotion as dynamic, culturally mediated, and socially constructed in interaction [2]. Their account shifts the goal of affective systems from identifying the correct inner state toward supporting interpretation, experience, and reflection.

This shift matters for wearable expression. A wearer may wish to communicate uncertainty, soften a statement, invite conversation, or leave a feeling unnamed. A system that infers and announces a category can undermine those intentions. Howell et al.'s study of real-time emotional biosensing similarly demonstrates tensions when physiological data acquire apparent authority over lived experience [17]. Stitching Affective Light therefore privileges authored expression. A future sensor may suggest or modulate a state, but the method requires that a person can accept, reinterpret, override, dim, or switch off the signal.

2.2 Co-Design Through Making

Co-design relocates people from the role of informants toward that of partners in creative decision-making [3]. Sanders and Stappers distinguish probes, generative toolkits, and prototypes as different ways of making in co-design: probes open questions, toolkits help participants express and generate possibilities, and prototypes make futures tangible enough to experience and contest [4]. Cultural probes similarly use evocative prompts to elicit situated and inspirational responses rather than standardized requirements [6].

Our method combines these functions without treating them as interchangeable. Emotion-story prompts operate as probes; motif, placement, color, and rhythm resources form a generative toolkit; and the illuminated embroidery becomes an experiential prototype. The sequence is important. If electronics are introduced first, participants may optimize what the hardware can display. If emotion labels are introduced first, they may reproduce the researcher's categories. We therefore begin with boundaries and personal situations, move to material metaphors, and introduce technical parameters only after an expressive direction exists.

Calling a process co-design also creates obligations. It is not enough to offer customization after core meanings have been fixed. Reviews of participatory design in HCI emphasize attention to mutual learning, political commitments, participant roles, and where decisions actually reside [5]. For Stitching Affective Light, participant authority includes the right to reject suggested mappings, create an idiosyncratic legend, refuse body placement, and decide that an emotion should remain unlit or undisclosed.

2.3 Craft, Embroidery, and E-Textile Participation

Crafting technology can diversify who makes electronics and how electronic objects are understood. Buechley and Perner-Wilson show that hybrid practices combining sewing, painting, carving, and electronics reconfigure both process and culture [9]. Sketch&Stitch supports iterative e-textile fabrication by letting people sketch art and circuitry directly on fabric [10]. Weaving Lighthouses and Stitching Stories demonstrates that participatory e-textile making can support personally authored objects and storytelling when workshop structures are made accessible [11].

These works motivate three choices. First, the embroidered image is not a decorative shell for an electronic display; it is a primary representational layer. Second, technical templates should reduce barriers without replacing participant

decisions. Third, making pace and motor or sensory access are part of the method, not logistical afterthoughts. Prior work on dynamic clothing also suggests that people may value textile expressiveness precisely because it does not resemble a screen attached to the body [12].

2.4 Color, Temporal Light, and Affective Materials

Color-emotion associations show patterned tendencies but do not form a universal dictionary. Wilms and Oberfeld found effects of hue, saturation, and brightness, with higher saturation and brightness generally increasing arousal and arousal increasing across tested hues toward red [13]. Jonauskaite et al.'s large cross-cultural study found both common patterns and effects of linguistic and geographic proximity [14]. Lin et al. further demonstrate that color interacts with tangible form and other material qualities in emotional interpretation [15].

Temporal behavior contributes another expressive dimension. Affective chronometry treats responsiveness, duration, rhythm, inertia, aliveness, and range as material resources for representing change in arousal [16]. LightSpeak indicates that viewers use color, blinking speed, and rhythm when interpreting minimalist LED expression, while also confusing some intended states [18]. Related design research organizes thermal affect through composable parameters rather than fixed emotion labels [19], while the Somatic Drawing Tool treats bodily and synaesthetic articulation as personally situated [20]. Together, these works support a design space, not a lookup table.

2.5 Research Gap

Prior work establishes participatory making, expressive e-textiles, affective material parameters, and the need to resist emotion-as-fact. What remains underarticulated is a practical method that connects these concerns from personal experience to a finished illuminated embroidery. Stitching Affective Light addresses that gap by specifying how representational authority, textile composition, programmable light, placement, and disclosure can be negotiated within one co-design process.

3 RESEARCH APPROACH

3.1 Artifact-Centered Research through Design

We used Research through Design (RtD) to construct an intermediate-level method through making, articulation, and reflection [7]. Method design and artifact design informed one another. The reviewed literature identified candidate design dimensions, including color, rhythm, material form, placement, and disclosure, that could be translated into workshop choices. Material exploration then clarified which choices could be separated, while the worked example exposed points requiring facilitation, technical translation, or ethical boundaries. The resulting knowledge sits between a universal theory of affective expression and a description of one artifact: it is a structured proposal that can guide future co-design sessions while remaining open to revision.

The evidence base comprises: (1) the literature-grounded rationale for co-design, affective ambiguity, color, temporality, and e-textiles; (2) the specified workshop materials and six-stage procedure; (3) the physical and computational architecture integrating ordinary embroidery, commercially sourced luminous line, and Arduino control; and (4) a designer walkthrough producing the RippleGlow example. We did not collect participant data or maintain a coded longitudinal autoethnographic diary. The analysis therefore concerns the coherence and implications of the method as designed, not participant experience.

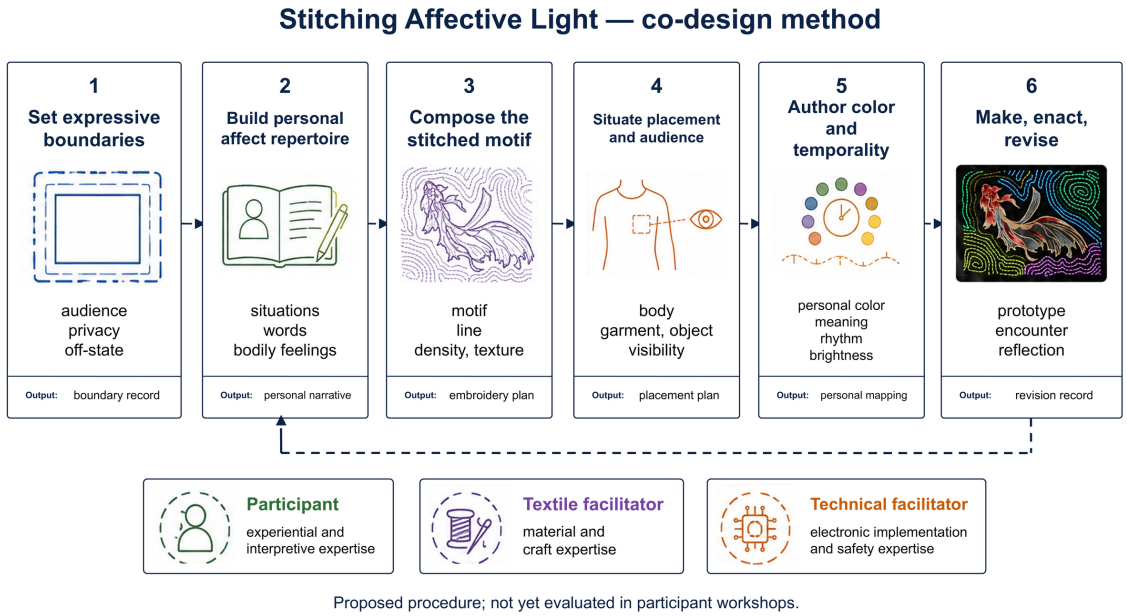


Fig. 1. Overview of the proposed Stitching Affective Light co-design method. Six stages move from expressive boundaries and a personal affect repertoire through motif composition, placement, color-temporality authoring, and iterative enactment. The lower row distinguishes participant, textile-facilitation, and technical-facilitation expertise. The procedure has not yet been evaluated in participant workshops.

3.2 From Artifact Dimensions to Method Stages

We began by separating four variables that were collapsed in the first artifact account: what is depicted, where it is encountered, how it is colored, and how it changes over time. We then added two cross-cutting requirements. Expressive boundaries must precede making because emotional visibility can be unwanted. Material enactment and reflection must close the loop because an imagined mapping may feel different when stitched, illuminated, and viewed at conversational distance. These decisions yielded six stages. Figure 1 summarizes the stage sequence and the distinct expertise brought by the participant, textile facilitator, and technical facilitator.

Table 1. Stitching Affective Light method. Stages describe a proposed co-design procedure; they have not yet been validated in participant workshops.

Stage	Purpose	Participant decision	Facilitator responsibility	Material output
1. Set expressive boundaries	Define what may be shown, to whom, and when.	Choose disclosure limits, opt-out conditions, and unwanted interpretations.	Establish consent, privacy, photosensitivity, and the right not to explain.	Boundary card and audience map.

Stage	Purpose	Participant decision	Facilitator responsibility	Material output
2. Build a personal affect repertoire	Move from generic labels to situated feelings and episodes.	Select words, metaphors, tensions, and states worth expressing.	Prompt without diagnosing or forcing disclosure.	Personal affect cards or short narratives.
3. Compose the stitched motif	Give affect a persistent material form.	Draw or assemble motif, density, line quality, and texture.	Offer accessible templates and technical stitch guidance.	Embroidery plan and unlit composition.
4. Situate placement and audience	Connect visibility to social context.	Choose chest, sleeve, shoulder, bag, object, or no body placement.	Explore viewing angle, comfort, bystanders, and reversibility.	Placement scenario and visibility plan.
5. Author color and temporality	Create a personal luminous vocabulary.	Assign, combine, or reject colors and rhythms.	Demonstrate parameters as options, not fixed meanings.	Personal mapping legend and light states.
6. Make, enact, and revise	Test how the expression feels in material and social use.	Revise motif, mapping, salience, or disclosure rules.	Implement safely, expose mapping controls, and invite interpretation without grading correctness.	Illuminated embroidery prototype, participant-editable mapping configuration, and revision record.

3.3 Analytic Questions

For each stage, we examined four questions: Who makes the consequential decision? What tacit or experiential knowledge does the activity seek to surface? What does the material make easy or difficult to express? What new obligation does the resulting visibility create? We then traced these questions through the RippleGlow walkthrough. The analysis generated five propositions presented later in the paper. They are artifact- and method-grounded propositions, not themes derived from participants.

3.4 Researcher Position and Ethics

The method was designed by researchers with particular aesthetic and technical preferences. The worked example reflects the maker's familiarity with embroidery, attraction to aquatic imagery, and access to Arduino components. These conditions can make the process appear more intuitive than it may be for a novice. No participant or personal data were collected. The method is not a diagnostic or therapeutic intervention, and future work with autistic people, people experiencing social anxiety, or other communities must begin with consultation rather than assuming that greater emotional visibility is desirable.

4 THE STITCHING AFFECTIVE LIGHT CO-DESIGN METHOD

4.1 Method Logic: From Experience to Material Expression

The method translates an affective intention through a chain of authored transformations: situated experience → personal vocabulary → textile metaphor → placement and audience → color and temporal behavior → material

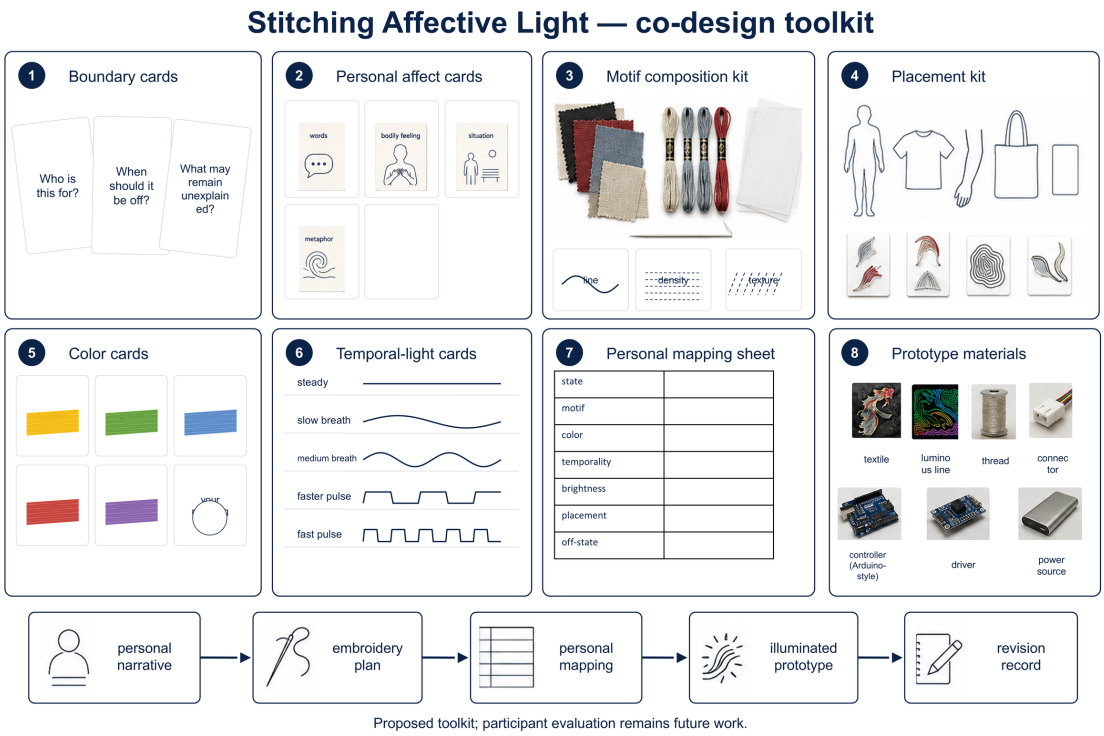


Fig. 2. Proposed co-design toolkit. Boundary cards, personal affect cards, motif and placement resources, color and temporal-light cards, a personal mapping sheet, and prototype materials support movement from a personal narrative to an embroidery plan, an illuminated prototype, and a revision record. The toolkit remains a design proposal pending participant evaluation.

enactment and revision. The chain is not a classifier pipeline. Each transformation introduces interpretation, and the participant can move backward, skip a representational layer, or preserve uncertainty. The facilitator's task is to keep possibilities open while making technical consequences understandable.

4.2 Stage 1: Set Expressive Boundaries

A session begins with boundaries rather than emotion labels. Prospective participants identify contexts in which they would or would not want a signal to be visible, people who may see it, and actions they need in order to withdraw. Prompt cards include: Who is this for? Who is it not for? What may remain ambiguous? When should the piece be dim, covered, detached, or switched off? The output is not a disclosure contract; it is a revisable design constraint that follows the artifact through later stages.

This stage prevents visibility from being treated as automatically beneficial. It also broadens the design space beyond clothing. A participant may select a bag, desk object, room textile, or temporary hand-held display if body placement feels too intimate. A co-design method for affective wearables must make non-wear and non-disclosure legitimate outcomes.

4.3 Stage 2: Build a Personal Affect Repertoire

Instead of choosing from a universal emotion wheel, participants assemble a small repertoire of states that matter in their own lives. The facilitator can offer valence and arousal as optional coordinates [1], but should also invite mixtures, transitions, bodily sensations, metaphors, and relational situations. A state might be described as 'quiet but expectant,' 'pressured on the outside and steady inside,' or 'wanting company without conversation.' Such descriptions provide richer design material than a single category.

The activity can use words, drawing, collage, objects, or gesture. Participants decide which parts are recorded. The method does not require disclosure of causes or histories. Its aim is to create an expressive repertoire, not to elicit clinical testimony.

4.4 Stage 3: Compose the Stitched Motif

Participants translate one or more affective descriptions into an embroidered composition. They choose figurative or abstract motifs, line direction, density, scale, texture, negative space, and which elements should remain constant. A template library can lower the threshold to participation, but blank fabric and free drawing must remain available so that the library does not become a hidden codebook.

The stitched motif is designed as a persistent layer: it remains visible when the piece is unpowered. Participants are therefore asked to consider what the artifact communicates while silent. This question distinguishes illuminated embroidery from a screen whose content can disappear completely. It also recognizes that time, effort, and handwork may become part of the meaning of the piece [9].

4.5 Stage 4: Situate Placement and Audience

Participants place paper or fabric mock-ups on a garment, accessory, or nearby object and observe them at conversational distance. We use face-to-face conversation as the primary design scenario because emotion is interpreted there through a changing combination of speech, silence, facial expression, gesture, clothing, relationship, and setting. The illuminated embroidery can therefore be studied as one supplementary cue within interaction rather than as a stand-alone label. Conversation also makes misunderstanding, repair, selective explanation, and withdrawal observable. It is an analytical scenario, not the only intended use context. Placement is discussed as an interaction variable: chest placement can be conspicuous and tightly associated with the self; a sleeve or shoulder may be peripheral; a bag can create distance; and an interior lining can allow selective reveal. The activity includes imagined bystanders and unintended audiences, not only an intended conversation partner.

This stage yields a visibility plan that records orientation, distance, attachment, removal, and a fallback unlit state. Comfort, washability, and electrical safety require separate technical evaluation before prolonged wear.

4.6 Stage 5: Author Color and Temporal Behavior

The facilitator demonstrates colors and temporal behaviors as manipulable materials. Literature-informed starting points can help participants begin, but no assignment is treated as correct. Participants can keep, invert, combine, or reject suggestions and record their own legend. Color primarily varies affective quality and broad valence-arousal region; temporal behavior varies persistence, fluctuation, and relative activation. The distinction reduces the pressure on a single visual variable to carry an entire emotion. Figure 3 makes this separation explicit by combining four expression dimensions with a provisional color-temporality matrix and documentary prototype states.

Table 2. Initial color resources offered to participants. These are prompts for reassignment, not universal emotion labels.

Color	Literature-informed starting region	Possible qualities	Co-design instruction
Yellow	Positive, higher activation	Happy, excited, expectant, lively	Keep, redefine, mix, or reject.
Green	Positive, lower activation	Relaxed, safe, relieved, satisfied	Ask whether cultural or personal meanings differ.
Blue	Lower activation	Calm, still, reflective, tired, or low	Do not force a positive or negative valence.
Red	Higher activation	Angry, anxious, tense, passionate, or excited	Treat as activation-capable, not as 'anger.'
Purple	Mixed or deliberately ambiguous	Contradictory, unresolved, difficult to classify	Use as an authored opening, not an evidence-based universal.

Table 3. Initial temporal-light resources. Rates are prototype parameters chosen for contrast; they are not psychophysical thresholds.

Mode	Nominal rate	Temporal quality	Co-design question
Steady	0 Hz	Sustained presence	Should this state remain stable, or does stability feel too definitive?
Slow breath	~0.1 Hz	Gradual expansion and return	Does the ten-second cycle feel calm, heavy, distant, or something else?
Medium breath	~0.25 Hz	Noticeable fluctuation	Does the four-second cycle fit the participant's sense of change?
Faster pulse	~0.5 Hz	Sharper activation	Is the signal expressive or already too attention-seeking?
Fast pulse	~1 Hz	Urgent, highly salient change	Should this option be softened or excluded for comfort and photosensitivity?

4.7 Stage 6: Make, Enact, and Revise

The participant and facilitator produce a low- or medium-fidelity illuminated embroidery, preserving a visible record of the participant's choices. The participant then encounters the piece unlit, illuminated, attached, and viewed from a distance. Reflection prompts ask: Does the light dominate the stitch? Does the rhythm feel like the intended state? Is the signal too legible, too ambiguous, or too public? What would you change or choose not to show?

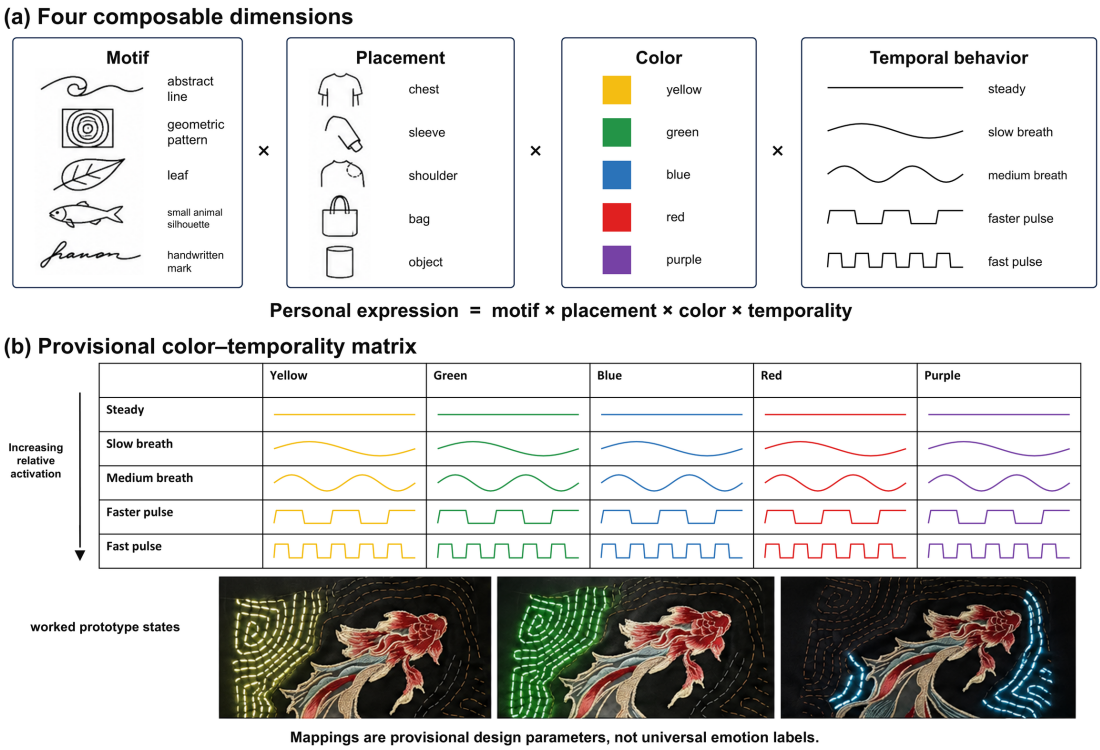


Fig. 3. Composable dimensions and provisional color-temporality matrix. Motif and placement shape persistent and situated meaning, while color and temporal behavior shape the luminous layer. The three documentary prototype states illustrate combinations rather than validated or universal emotion labels; participants may redefine, combine, or reject the proposed mappings.

Enactment is not a recognition test. Another person may describe what they notice, but their interpretation is not graded against an emotion label. Differences become material for revision and negotiation. The method records changes to the personal legend, motif, placement, brightness, and off-state so that interpretation remains revisable rather than locked into the first design decision.

4.8 Toolkit Components and Facilitation

Table 4. Proposed co-design toolkit and the design work each component supports.

Component	Contents	Purpose	Avoid
Boundary cards	Audience, context, opt-out, concealment, off-state prompts	Make autonomy and selective disclosure designable.	Assuming visibility is desirable.
Affect prompts	Words, blank cards, metaphors, body/temporal prompts	Support personal repertoires without diagnosis.	Forcing a standard emotion list.

Component	Contents	Purpose	Avoid
Motif kit	Fabric shapes, tracing paper, threads, pens, texture samples	Move from felt experience to persistent composition.	Presenting templates as correct answers.
Placement kit	Body/garment/object silhouettes and removable mock-ups	Explore situated visibility and audience.	Restricting outcomes to chest-worn badges.
Light cards	Five colors, waveform cards, brightness examples	Make computational parameters tangible and comparable.	Equating one color with one emotion.
Mapping sheet	State, motif, color, rhythm, placement, off-state	Preserve participant-authored logic for implementation.	Letting a programmer silently reinterpret choices or requiring code edits for every revision.
Prototype kit	Embroidery substrate, thread, luminous line, controller, power	Materialize and revise the expression through reversible connectors and a pretested control module.	Hiding safety or material limitations, or expecting every participant to fabricate the circuit.

Figure 2 assembles the proposed materials into an eight-part toolkit and shows the intended progression from personal narrative to revision record. Facilitation should preserve three distinct forms of expertise. Participants hold experiential and interpretive expertise; textile facilitators contribute knowledge of stitch, tension, density, and fabrication; technical facilitators contribute knowledge of power, control, attachment, and safe operating limits. Co-design occurs when these forms of expertise negotiate the outcome, not when technical expertise merely implements a designer-authored concept. A participant does not need to master every stitch or circuit to remain an author, but must understand the available options, approve consequential translations, and be able to revise the personal mapping without having to renegotiate its meaning through a programmer.

We therefore organize participation across three linked layers. The participant-authored layer contains the motif, placement, personal color and temporal meanings, brightness, and disclosure rules. The co-made assembly layer covers routing the luminous line, choosing attachment points, testing flexibility, and connecting a prepared module to the textile. The specialist infrastructure layer contains the driver circuit, power protection, firmware, and safe operating limits. This layered arrangement lets participants combine textile and computational elements through consequential design decisions and supported assembly without requiring them to become electronics specialists. Whether the arrangement works for novices remains an empirical question for participant workshops.

5 MATERIAL AND COMPUTATIONAL IMPLEMENTATION

5.1 Persistent Textile Layer

The method does not prescribe a motif. Its material requirement is that participants can compose a readable or meaningful unlit textile state using conventional embroidery or another compatible stitch practice. Thread color, line density,

direction, texture, and negative space are expressive parameters. The luminous element may follow, border, cross, or interrupt stitched forms, but should not erase the authored textile composition when power is absent.

5.2 Transient Luminous Layer

The current implementation uses a commercially purchased luminous line integrated with the embroidery and controlled through Arduino. The luminous material itself is not our invention. The contribution lies in how the method connects participant-authored meanings to controllable color and temporality.

The minimum control architecture is: manually selected expressive state → personal mapping table → Arduino-based controller and appropriate driver → color and temporal behavior → luminous line. Manual state selection preserves authorship and makes mappings inspectable. In the current RippleGlow prototype, however, a participant can select among programmed states but cannot independently change which color or temporal pattern belongs to a state; remapping requires access to the Arduino configuration and technical support. We report this as a limitation of the current implementation rather than as completed end-user control.

5.3 Participant Editable Mapping and Layered Technical Participation

The final prototype should expose the personal mapping table as an end-user control rather than hide it in firmware. A simple on-device or local companion editor should let participants assign a color, temporal mode, brightness, and off-state to each personally named state; preview the result; and save or restore mappings without editing code or reflashing the controller. The configuration should be stored separately from the control program so that technical maintenance cannot silently overwrite personal meanings. A physical switch-off control must remain available even if the editor is unavailable.

Textile and technical integration should likewise be modular. Participants can stitch or direct the visual composition and route luminous line using marked channels, strain relief, and reversible connectors, while a pretested controller-driver-power module contains safety-critical electronics. Facilitators explain material constraints and offer alternatives, but the participant decides which compromises are acceptable. This approach combines textile and professional technical expertise through a legible interface and negotiated handoff rather than expecting every participant to fabricate the circuit. Figure 4 shows how these authored decisions connect to the modular technical stack while keeping current limitations and next-iteration controls visible.

5.4 Technical Documentation Required for Reproduction

A reproducible implementation must document dimensions, substrate, stitch types, attachment method, luminous-line construction, driver circuit, supply voltage and current, waveform definition, brightness range, timing accuracy, thermal behavior, bending radius, runtime, failure modes, washability, connector design, and the format and editing procedure for personal mappings. These measurements were not retained under controlled conditions for the present artifact, and an end-user mapping editor has not yet been implemented. The implementation should therefore be understood as an early functional pattern, not a production-ready wearable.

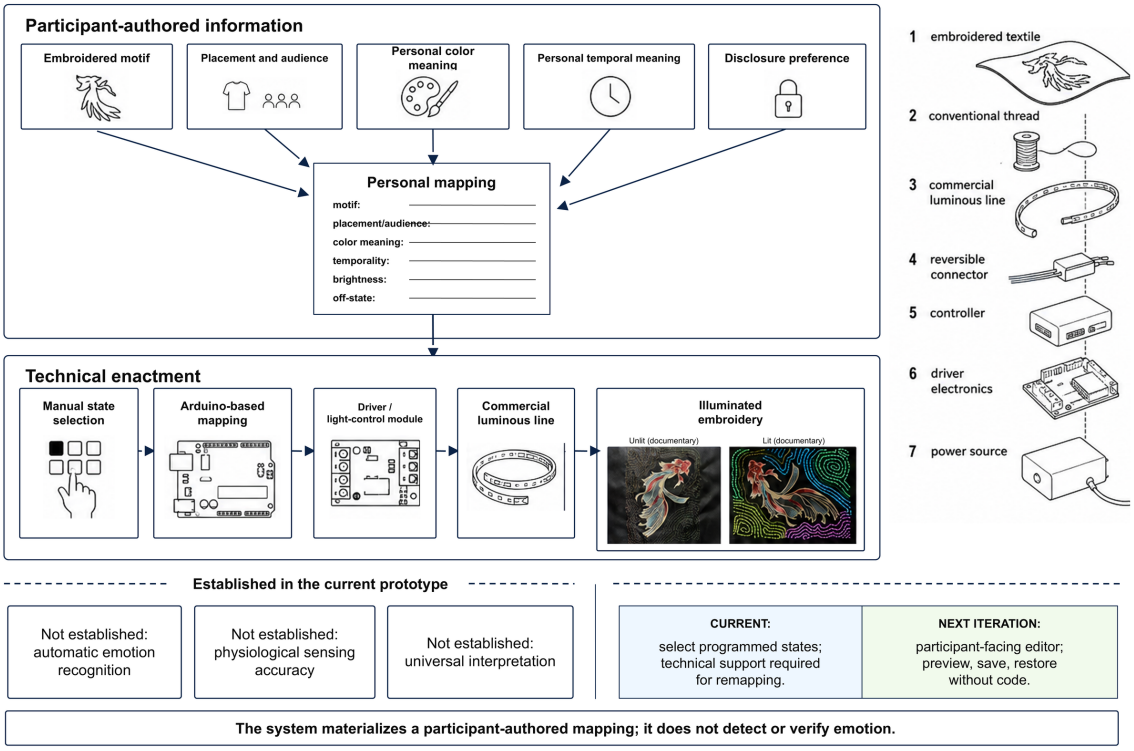


Fig. 4. Translation from participant-authored information to technical enactment. A personal mapping sheet connects motif, placement and audience, color, temporality, and disclosure preferences to manual state selection, Arduino-based mapping, a driver module, commercial luminous line, and illuminated embroidery. The current prototype requires technical support for remapping; a participant-facing editor remains a next-iteration requirement. The system does not automatically detect, classify, or verify emotion.

6 WORKED EXAMPLE: RIPPLEGLOW

6.1 Why a Worked Example

RippleGlow is a researcher-authored worked example that demonstrates how the method's decisions can converge into one artifact. It does not define what future participants should make. A worked example is useful because it exposes relationships that remain abstract in a toolkit specification: whether a stable motif and dynamic field can coexist, how luminous lines participate in composition, and how a personal mapping can remain legible without becoming universal. Documentary states of this worked example are included in Figures 3 and 4 to connect the abstract design space and system architecture to the constructed artifact.

6.2 From Personal Metaphor to Composition

The example begins from the metaphor of a relatively stable self, situated within a changing emotional environment. This became a single goldfish surrounded by sparse, linear marks representing an ocean. The fish is the persistent focal motif; the ocean is the principal luminous field. Excluding plants, bubbles, and detailed scenery keeps the composition open enough for color and rhythm to change its affective reading.

In the method's terms, this is one answer to Stage 3, not a template. Another person might embroider a knot, a house, a cloud, a repeated geometric field, handwriting, or no figurative object at all. What transfers is the question of which elements should remain stable and which should become dynamically illuminated.

6.3 Mapping Color and Rhythm

The example uses yellow, green, blue, red, and purple as starting colors and combines them with steady, slow-breathing, medium-breathing, faster-pulse, and fast-pulse states. Yellow can support positive and activated qualities; green can support positive lower activation; blue can support calm, still, reflective, or low states; red can support high activation across negative and positive valence; and purple remains intentionally available for mixture or contradiction. These meanings remain the maker's provisional legend.

Separating color from temporality enables difference without multiplying literal labels. Red with slow breathing may suggest sustained tension, while red with rapid pulses may suggest acute activation. Blue with steady light may suggest calm continuity, while blue with medium breathing may suggest a low but shifting state. The same combination may be reassigned by another person.

6.4 Placement and Situated Use

RippleGlow is small enough to be worn on the chest, sleeve, or shoulder, or attached to a bag. The method treats these options as different disclosure arrangements. Face-to-face conversation is used because it allows the light to be interpreted alongside speech, facial expression, gesture, relationship, and setting, and because disagreements can be repaired through clarification or withdrawal. The light supplements rather than replaces these cues. The wearer may revise, dim, switch off, conceal, or leave the signal unexplained.

7 METHOD- AND ARTIFACT-CENTERED ANALYSIS

7.1 Proposition 1: Co-Design Must Transfer Interpretive Authority

The strongest distinction between this method and customization is where meaning is decided. Selecting among designer-authored badges changes appearance while leaving the representational system intact. Stitching Affective Light asks the participant to author both the visible form and the mapping logic. A participant can decide that red means safety, that no color represents sadness, or that a particular state should only be expressed through stitch density. The method's claim to co-design depends on preserving this authority in facilitation and implementation.

This proposition also changes documentation. The personal mapping sheet is not a temporary workshop artifact to be translated invisibly by a designer; it is part of the system specification. Any technical simplification that changes its meaning should be returned for negotiation.

7.2 Proposition 2: Persistent and Transient Layers Support Different Kinds of Expression

Embroidery persists across time and power states, while light can appear, change, and disappear. The persistent layer can carry identity, metaphor, and craftsmanship; the transient layer can carry activation, fluctuation, and situational emphasis. This division gives participants more than a larger palette. It lets them decide which meanings should remain visible and which should be temporary.

The distinction is not absolute. An unlit motif already affects interpretation, and repeated light behavior may become socially associated with the wearer. Facilitators must therefore discuss the meanings of both layers and their interaction rather than treating the fabric as neutral hardware support.

7.3 Proposition 3: Ambiguity Should Be Authored, Not Merely Tolerated

Ambiguity can invite curiosity, appropriation, and multiple readings [8], but it can also create misunderstanding. The method makes ambiguity an explicit participant decision. A figurative motif, a broad color region, a private legend, a partially hidden placement, or an off-state can each influence how much information is offered for interpretation. The design goal is not maximum legibility. It is a participant-controlled relation between legibility and openness.

7.4 Proposition 4: Material Access Shapes Participation

Embroidery can be slow, technically demanding, visually inaccessible, or physically difficult. Electronics introduce further asymmetries in confidence and expertise. A method cannot claim inclusive co-design merely because it uses craft. Accessible participation may require fabric or templates with stitch holes prepared in advance, larger-scale materials, tactile or high-contrast templates, alternative fastening, machine embroidery, verbal composition, paired making, or participant direction without manual stitching. The relevant measure is control over consequential decisions, not whether every person performs every fabrication step.

7.5 Proposition 5: Emotional Visibility Requires Reversible Disclosure

Placement, brightness, and temporal salience shape who notices a signal, but visibility can exceed the participant's intention. We focus on conversation because the signal enters an already active field of speech, gesture, facial expression, relationship, and context: a partner can ask, misread, or repair an interpretation, while bystanders may still see a signal not meant for them. Rapid pulses may draw attention, and an observer may privilege the light over the wearer's words. The method therefore treats off, dim, cover, detach, and no-explanation options as core interaction states. Privacy is not only a data policy; it is a material and social property of the artifact.

8 DISCUSSION

8.1 What the Method Adds to Affective Interface Design

Stitching Affective Light reframes affective interface design from choosing how a system should represent emotion to arranging how people can author representations with material and computational support. This reframing makes personal meaning, not automatic inference, the starting point. It also provides a concrete route from open-ended affective experience to implementable parameters without claiming that the parameters are emotionally universal.

The method's four-part expressive grammar—motif, placement, color, and temporality—supports modularity without reducing expression to interchangeable settings. Each dimension answers a different question: What remains materially present? Where and to whom is it visible? What affective quality is suggested? How does that quality persist or change? Their combination creates a design language that can be shared procedurally while remaining personal semantically.

8.2 Co-Design Is a Future Empirical Commitment

The current work designs for co-design but has not yet demonstrated co-design in practice. This distinction is not a minor limitation. Participation will reshape the method: prompts may be intrusive, stages may occur in a different order,

the five-color palette may be constraining, and embroidery may not be the preferred medium. The current prototype also requires technical support to change mappings. A participant study must therefore evaluate not only whether people can complete the activities, but whether they experience meaningful authorship, can modify mappings without a facilitator, can understand the textile-electronic boundary, and can contest the research premise [5].

For CHI, the value of the present paper is consequently methodological and generative. It specifies a design object that future research can test rather than reporting a polished toolkit as already validated. The worked example functions as a walkthrough and boundary object: it makes the method discussable while remaining explicitly partial.

8.3 Implications for Future Affective Wearables

First, systems should expose personal mappings through controls that participants can revise without editing firmware or relying on a facilitator. Second, the mapping configuration should remain separate from safety-critical electronics so that personal meaning and professional technical maintenance can coexist. Third, designers should give the unpowered state equal attention because it remains socially and materially active. Fourth, evaluation should include ambiguity, contradiction, repair, and refusal, not only recognition accuracy. Fifth, sensing—if introduced—should produce suggestions with uncertainty and override rather than authoritative declarations. Finally, expressive wearables should be evaluated within cue ecologies that include conversation, gesture, clothing, relationship, and bystanders.

8.4 A Staged Evaluation Program

We propose four subsequent studies. Study 1 should run formative workshops with a small, diverse adult sample to examine stage order, prompt clarity, material accessibility, authorship, and facilitator intervention. Study 2 should technically characterize brightness, timing, power, heat, bending, attachment, connectors, and failure modes while implementing the participant-facing mapping editor. Study 3 should test whether participants can remember, independently revise, and restore their mappings, and whether modular textile-electronic assembly supports meaningful participation without requiring circuit expertise. Study 4 should use dyadic, face-to-face scenarios to investigate how illuminated embroidery interacts with speech, gesture, disagreement, repair, and selective disclosure.

Work with specific communities should follow community consultation and ethics review. Research with autistic people or people experiencing social anxiety must not assume that more visible emotion is beneficial. It should examine sensory load, stigma, masking, forced disclosure, cultural expectations, and the option to reject the device or redefine the research goal.

9 CONCLUSION

Stitching Affective Light is a six-stage method for co-designing personal illuminated embroideries. It begins with expressive boundaries, builds a situated affect repertoire, develops a persistent stitched motif, situates placement and audience, authors color-temporality mappings, and closes through making, enactment, and revision. The method treats emotion as interpreted and negotiated rather than automatically revealed.

RippleGlow—a goldfish within a color-changing linear ocean—demonstrates one passage through the method. Its value is exemplary rather than prescriptive. By separating motif, placement, color, and temporality, the paper offers a reusable grammar for personal expression; by distinguishing participant expertise from textile and technical expertise, it specifies where co-design authority should reside. The current prototype still requires technical support for remapping, so independent mapping control and modular textile-electronic assembly remain explicit requirements for the next

iteration. The next step is not to generalize from the goldfish, but to invite diverse participants to challenge, reshape, and potentially refuse the method.

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